

Rover Rescue: Plan a Fair Test

I can plan a fair investigation of how surface type affects a rover's travel distance.

CHANGE

use evidence

MEASURE

use evidence

KEEP

use evidence

[Open the original interactive lesson](#)

What success looks like

- I can name surface type as the independent variable and compare smooth with fabric.
- I can measure travel distance and keep other variables controlled.
- I can plan three repeats for each condition and state a safe method.

Retrieve and reconnect

- Name the independent variable in a surface test.
- Name the dependent variable and its unit.
- List two variables to control.
- Explain why each condition needs repeats.

Which plan lets us find out whether surface type affects rover travel distance?

Think first. Point, say, sign or write your choice. Give one reason.

- A. Change surface only; keep ramp and release the same
- B. Change surface and ramp height together
- C. Push the rover differently each time

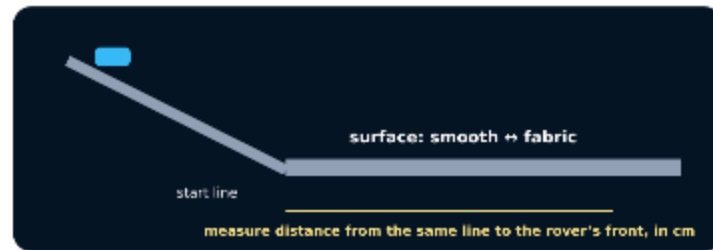
Look closely · what is the evidence?

CHANGE · one independent variable

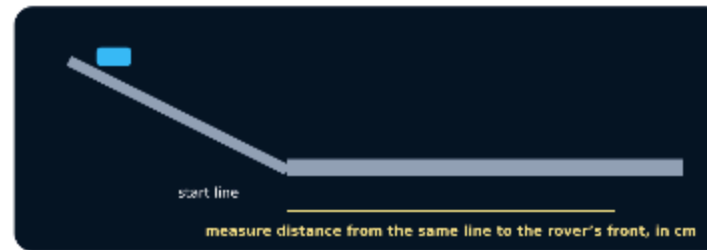
MEASURE · distance in centimetres

KEEP · controls and repeat

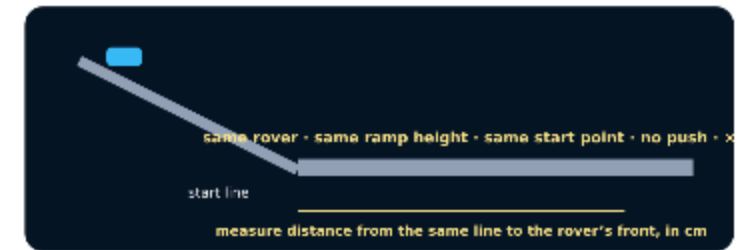
Look closely · what is the evidence?



Ramp and surface



Measuring distance



Controls

Words we will use

- independent variable — what I change
- dependent variable — what I measure
- control variable — what I keep the same

Watch the model

- Choose one question: How does surface type affect rover travel distance? Conditions: smooth and fabric.
- Measure the distance from the same finish line to the rover's front in centimetres.
- Keep the rover, low ramp height, start point and no-push release the same; plan three repeats per condition.

Which is the fairest rover test question?

- A. How does surface type affect travel distance when ramp height and release stay the same?
- B. What happens when surface and ramp height both change?
- C. Which rover looks fastest after a different push each time?

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Safety first: use a stable low ramp in a clear tray or lane. Release without pushing and keep hands out of the travel path.
- Create a six-row table before testing: two conditions with three blank repeat rows each.
- Make a prediction with a reason, but record what happens rather than changing results to fit it.

Find and repair the misconception

For a clearer result, change the surface and ramp height together.

A. Keep it: two changes give twice as much evidence.

B. Repair it: change one variable so the cause of any difference is clearer.

C. Change it: add a different push on every run.

Fair-Test Control Board

Place each method card under Change, Measure or Keep the same. Reject any plan that changes two variables.

- Choose one question
- Name two conditions
- Set measure and controls
- Plan three repeats

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- CHANGE
- MEASURE
- KEEP THE SAME
- REPEAT / RECORD

Read the frozen plan as a method: We change surface. We measure _____. We keep _____ the same. We repeat _____.

Your independent mission

Produce a safe fair-test plan comparing smooth and fabric surfaces, with three repeats per condition.

Record: Question, prediction, variable table, safe numbered method, two conditions, six result spaces and one model limitation.

Choose your support and challenge

- Supported: Use the prepared surface question and complete Change, Measure, Keep and Repeat with icon-backed cards.
- Standard: Write a fair question, prediction, variable table and numbered method for two conditions with three repeats.
- Stretch: Justify the controls, predict random variation and explain one limitation of transfer to a real rover.

Show what you know

Name the dependent variable and two controls.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.